

LESSON 2

MEASURES OF DISPERSION

Overview:

In the preceding lesson, measures of central tendency such as the mean, median, and mode of a data set were introduced. Some characteristics of a set of data may not be evident from an examination of mean, median, or mode. This lesson will show that it is not enough to get measures of central tendency in a data set. The range, standard deviation, and variance will be discussed in this lesson.

Learning Outcomes:

At the end of Lesson 2, the students can:

1. discuss the need for analyzing data using the different measures of dispersion;
2. differentiate the different measures of dispersion; and
3. provide a sound interpretation of these measures.

Duration: 1.5 hour

Materials Needed:

Chalk-board, Slide Presentation, and Handouts

Learning Contents:

2.1 THE RANGE

The **range** (R) of a set of data values is the difference between the highest data value (H) and lowest data value (L). In symbols, $R = H - L$.

Example

A testing lab wishes to test two experimental brands of outdoor paint to see how long each will last before fading. The testing lab makes 6 gallons of each paint to test. Since different chemical agents are added to each group and only six cans are involved, these two groups constitute two small populations. The results (in months) are shown. Find the range of each brand.

Brand A	Brand B
10	35
60	45
50	30
30	35
40	40
20	25

Solution

Brand A, the range is

$$R = H - L = 60 - 10 = 50 \text{ months}$$

Brand B, the range is

$$R = H - L = 45 - 25 = 20 \text{ months}$$

2.2 THE STANDARD DEVIATION

The **standard deviation** of a set of numerical data makes use of the amount by which each individual data value deviates from the mean. These deviations, represented by $(x - \bar{x})$, are positive when the data value x is greater than the mean $\bar{x}$ and are negative when x is less than the mean $\bar{x}$. The sum of all the deviations $(x - \bar{x})$ is 0 for all sets of data. Because the sum of all the deviations of the data values from the mean is always 0, we cannot use the sum of the deviations as a measure of dispersion for a set of data. Instead, the standard deviation uses the sum of the squares of the deviations.

Standard Deviations for Populations and Samples

If $x_1, x_2, x_3, \dots, x_n$ is a population of n numbers with a mean of μ , then the standard deviation of the population is $\sigma = \sqrt{\frac{\sum(x-\mu)^2}{n}}$.

If $x_1, x_2, x_3, \dots, x_n$ is a sample of n numbers with a mean of $\bar{x}$, then the standard deviation of the sample is $s = \sqrt{\frac{\sum(x-\bar{x})^2}{n-1}}$.

Steps in Computing the Standard Deviation (for a sample)

- Step 1: Compute the mean if it is unknown.
- Step 2: List all the data values, x in a column. This will be the first column.
- Step 3: Provide the second column by subtracting the mean from each data Value $(x - \bar{x})$.
- Step 4: In the third column square each of the results from step 2, $(x - \bar{x})^2$.

- Step 5: Add all the results in column 3, $\sum(x - \bar{x})^2$.
- Step 6: Substitute all the necessary information to the formula. Be careful to check whether the data is taken from a sample or population.

Example

Let us use the previous example on the range. Suppose we want to find the standard deviation of Brand A.

Step 1 Since the mean is needed in computing the standard deviation, we have to find the mean first. That is,

$$\bar{x} = \frac{10 + 60 + 50 + 30 + 40 + 20}{6} = \frac{210}{6} = 35$$

Step 2	Step 3	Step 4	
Brand A (x)	$x - \bar{x}$	$(x - \bar{x})^2$	
10	$10 - 35 = -25$	$(-25)^2 = 625$	
60	$60 - 35 = 25$	$25^2 = 625$	
50	$50 - 35 = 15$	$15^2 = 225$	
30	$30 - 35 = -5$	$(-5)^2 = 25$	
40	$40 - 35 = 5$	$5^2 = 25$	
20	$20 - 35 = -15$	$(-15)^2 = 225$	
$\sum(x - \bar{x})^2 = 1750$			← Step 5

Step 6 Since the data is taken from a sample, then we will use $s = \sqrt{\frac{\sum(x-\bar{x})^2}{n-1}}$, where $n = 6$ and $\sum(x - \bar{x})^2 = 1750$.

Thus,

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}} = \sqrt{\frac{1750}{6 - 1}} = \sqrt{\frac{1750}{5}} = \sqrt{350} \approx 18.71$$

Hence, the given set of data differs from each other represented by the mean of the data at an average of 18.71 months or almost 19 months.

2.3 THE VARIANCE

Variance is the average of the squares of the distance that each value is from the mean.

Variance for Populations and Samples

If $x_1, x_2, x_3, \dots, x_n$ is a population of n numbers with a mean of μ , then the variance of the population is $\sigma^2 = \frac{\sum(x-\mu)^2}{n}$.

If $x_1, x_2, x_3, \dots, x_n$ is a sample of n numbers with a mean of $\bar{x}$, then the variance of the sample is $s^2 = \frac{\sum(x-\bar{x})^2}{n-1}$.

Example

Let us say we want to know the variance of Brand A from the previous example.

We already know the standard deviation of the data. That is $s \approx 18.71$.

By squaring both sides, we get $s^2 \approx 18.71^2 \approx 350.06$.

Learning Activity:

Direction: This is a group activity with two or three members. Using the data in this link <https://drive.google.com/drive/folders/1oQxvjlsfDJHFAyqH8-kq2x5aRHRRXGJ?usp=sharing> find the **range**, **standard deviation**, and **variance** of the group's assigned variable. Provide a **sound interpretation** of the result.

- Group 1:** Age
- Group 2:** Height
- Group 3:** Weight
- Group 4:** Monthly Expenses
- Group 5:** Monthly Internet Allowance
- Group 6:** General Mathematics Final Grade
- Group 7:** Statistics and Probability Final Grade
- Group 8:** Daily Calorie Intake
- Group 9:** Blood Pressure (Systolic)
- Group 10:** Blood Pressure (Diastolic)
- Group 11:** Heart Rate
- Group 12:** Number of Hours Spent in Social Media

NOTE: You can also check your group in the given link above. Submit your work in the LMS in a single **JPEG** (picture) format. Since every group has two or three members, Student A will compute the range; Student B for standard deviation; Student C for the variance. In the case of a group with only two members, Student A and Student B will decide on what measures of dispersion to compute. This is a group activity but the submission shall be of **individual** basis.

Learning Evaluation:

1. The following table displays the mothers’ ages of the 15 GEC 104 students. Find the range, standard deviation, and variance of the data set.

36	44	47	52	45
40	39	51	58	51
39	42	50	36	38

2. Will it be possible for a data set to have a standard deviation of 0? Explain your answer.
3. An article recently reported that the average income of 40 college professors was ₱43,250 with a standard deviation of ₱1,250. What does this mean?

References:

Aufman, R. et. al. (2018). *Mathematics in the Modern World*. Rex Book Store Inc., Sampaloc, Manila, Philippines.

Blalock, H. Jr. (1960). *Social Statistics*. McGraw-Hill Book Company, Inc, United States of America

Brase,C. et.al. (2018). *Understanding Basic Statistics*. 8th Edition. Cengage Learning.